Audited Few-Shot Personalization, Leakage-Free Dataset Segregation, and Empirical Parity Analysis on the TORGO Corpus

Project Title: Low-Resource Personalized Atypical Speech Recognition Using Hybrid Quantum-Classical Deep Learning for Reliable Emergency Communication

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Target Dataset: TORGO Database of Acoustic and Articulatory Speech from Speakers with Dysarthria (15 Speakers: 8 Dysarthric, 7 Control)

Canonical Operational Classifier: qsvc_q8_r1 (8-Qubit Fidelity Quantum Kernel Support Vector Classifier)

Primary Matched Classical Comparator: Radial Basis Function Support Vector Machine (RBF-SVM, d=8, gamma=0.125, C=1.0)

2. PURPOSE OF THE NEW EVALUATION

The purpose of this resegregated evaluation is to establish an unassailable, leakage-free empirical benchmark comparing quantum-enhanced classifiers against matched classical baselines for personalized atypical speech recognition. While earlier exploratory phases in the project established preliminary feasibility, forensic audits of historical dataset splits revealed critical vulnerabilities—specifically, prompt text repetition between adaptation and test sets, potential cross-microphone contamination between array and head microphones, and shifting test set support across sample budgets. This report documents the complete resegregation protocol, executes paired model evaluations across 60 rigorously controlled speaker-by-budget conditions, and provides verifiable evidence to resolve the core scientific inquiry: whether parameter-free quantum kernels yield a demonstrable advantage over classical statistical learning under extreme low-resource constraints.

3. AUDIT OF THE PREVIOUS DATASET SPLIT

A forensic audit of the historical splits/torgo_splits.json file revealed that while recording IDs were disjoint, 553 prompt text types overlapped between calibration and evaluation subsets. In atypical speech processing, prompt repetition introduces a severe confound: acoustic classifiers can exploit lexical memorization of specific prompt phrases (such as repeated isolated words 'yes', 'no', 'help', or 'water') rather than generalizing to novel emergency utterances from the target speaker. Furthermore, an audit of the initial Phase 8 80/20 benchmark revealed that 218 test samples (17.94% of the 1,215 test cohort) had appeared in the historical Phase 6.5 few-shot adaptation pool used to fit earlier models, invalidating naive before-and-after longitudinal comparisons. These defects mandated the creation of a completely resegregated, versioned split manifest (splits/torgo_resegregated_v1.json) that enforces absolute prompt-group isolation and channel-stem disjointness.

4. FINAL DATASET SEGREGATION AND EVALUATION PROTOCOL

The resegregated evaluation protocol enforces strict scientific safeguards across all 15 TORGO speakers and four calibration budgets (N in {20, 30, 40, 50}):

1. Strict Nestedness: Calibration subsets follow exact inclusion hierarchy: S20 subset of S30 subset of S40 subset of S50, ensuring that budget expansion strictly adds information without altering earlier calibration samples.

2. Zero Cross-Microphone Leakage: Utterances are grouped by their physical speech event stem. If an utterance recorded via headMic is assigned to calibration, its corresponding arrayMic recording is permanently excluded from evaluation.

3. Zero Prompt Leakage: Transcripts normalized to alphanumeric stems are grouped into unique Prompt Group IDs (PGRPs). Calibration and evaluation partitions share exactly zero prompt groups.

4. Locked Evaluation Cohort: For each speaker, the evaluation test partition is fixed and identical across all sample budgets N (summing to exactly 5,317 test samples across the 15 speakers), isolating true sample efficiency from test-set composition shifts.

5. Upstream Preprocessing Fitting & Disclosure: Principal Component Analysis (PCA) projection matrices and StandardScaler transformations are fitted strictly on the target speaker's calibration partition (mean explained variance 83.48% +/- 3.12% across conditions, as documented in results/pca_explained_variance_by_condition.csv). Test samples are never seen during dimensionality reduction or threshold calibration. We explicitly disclose that all 60 conditions have zero validation samples; the operating threshold was fitted on calibration data, and metrics are reported on the strictly locked 5,317-item test set.

5. DATASET COMPOSITION AND THREE DISTINCT CLASS DISTRIBUTION PROFILES

To eliminate any ambiguity across artifacts, we explicitly distinguish the three different class distribution views present in the evaluation suite:

A. Full-Corpus Distribution (Table 5.1 & Figure 1): 16,552 total raw recordings across all 15 speakers, reflecting the 745:1 corpus-wide class imbalance.

B. Resegregated Test Evaluation Cohort (Table 5.2 & Figure 10/11): Summing the locked test IDs across all 15 speakers yields exactly 5,317 test samples: GENERAL_SPEECH (5,219; 98.16%), CRITICAL_RESPONSE (35; 0.66%), FALL (21; 0.39%), BREATHING_MEDICAL (18; 0.34%), PAIN (13; 0.24%), SOS_HELP (11; 0.21%), and LOCATION (0; 0.00%).

C. Per-Class and Confusion Matrix Supports: Evaluated strictly on the locked 5,317 test cohort, matching the evaluation manifest exactly.

Table 5.1: Class Composition: Full Corpus (16,552) vs. Locked Evaluation Test Cohort (5,317)

INTENT LABEL	CLINICAL / SEMANTIC INTENT	CORPUS COUNT	CORPUS PCT	TEST COHORT COUNT	TEST PCT

GENERAL_SPEECH	Non-emergency conversational & baseline speech	15,661	94.62%	5,219	98.16%
CRITICAL_RESPONSE	Immediate affirmations & negations ('Yes', 'No')	237	1.43%	35	0.66%
FALL	Physical distress, slips, falls ('I fell', 'Help me')	223	1.35%	21	0.39%
BREATHING_MEDICAL	Respiratory distress & choking ('Short of breath')	170	1.03%	18	0.34%
PAIN	Acute somatic pain ('Chest hurts', 'Severe pain')	136	0.82%	13	0.24%
SOS_HELP	Explicit distress screams & alarms ('SOS', 'Emergency')	104	0.63%	11	0.21%
LOCATION	Room positioning markers ('Bedroom', 'Kitchen')	21	0.13%	0	0.00% (Ultra-Sparse)
TOTAL EVALUATION	Audited TORGO Speech Cohort	16,552	100.00%	5,317	100.00%

6. MODEL INVENTORY AND EXPERIMENTAL CONFIGURATIONS

Ten computational models spanning quantum kernels, classical kernel baselines, neural adapters, and linear classifiers were evaluated under identical resegregated conditions:

1. QSVC Q8-R1 (Canonical Operational Model): 8-qubit quantum support vector classifier utilizing a second-order Pauli-Z feature map (ZZFeatureMap) with linear entanglement, circuit repetition depth R=1, and fidelity statevector transition kernel.

2. Matched Classical RBF-SVM (d=8): Non-linear support vector machine operating on the identical 8-dimensional PCA projection space with Gaussian radial basis function kernel K(x,z) = exp(-gamma ||x-z||^2), bandwidth gamma = 1/d = 0.125, and regularization C=1.0.

3. Linear SVM (d=8): Convex linear hyperplane classifier providing a linear geometric baseline in the 8-D PCA subspace.

4. K-Nearest Neighbors (KNN, k=5, d=8): Non-parametric instance-based metric learning baseline.

5. QSVC Q4-R1: Compact 4-qubit quantum kernel operating on the top 4 PCA components.

6. QSVC Q8-R2: Deeper 8-qubit quantum kernel with two feature map repetitions (R=2, circuit depth 18).

7. Phase 8 Track A: Independent classical residual module (384-D MLP -> scalar logit) fused with upstream emergency gate.

8. Phase 8 Track B: Independent 8-qubit variational quantum circuit (VQC) with 16 trainable parameters acting as a residual modulation module.

9. Phase 8 Track C: Matched classical shadow residual module (8-D MLP -> scalar logit) receiving identical input and optimization structure.

7. EVALUATION METRICS AND DUAL ESTIMAND REPORTING

Due to severe class imbalance (98.16% non-emergency speech in the test set), standard accuracy is uninformative. Performance is assessed using class-sensitive metrics: Macro-F1 across all 7 classes (with zero_division=0), Supported 6-Class Macro-F1 (excluding LOCATION), Balanced Accuracy (macro-recall across classes), Emergency Recall (TP / (TP+FN)), Emergency Precision (TP / (TP+FP)), False Emergency Rate (FER = FP / (FP+TN)), and Intent Preservation Error Rate (IPER = 1 - Emergency Recall).

Crucially, to resolve past reporting discrepancies, we explicitly report both statistical estimands: (1) Condition-Level Mean (unweighted average across the 60 speaker-by-budget conditions) and (2) Pooled Test-Set Metric (calculated from pooled predictions over the locked 5,317 test samples). Both are documented in Table 8.1.

8. CONSOLIDATED MASTER MODEL COMPARISON TABLE

Table 8.1 presents the master benchmark results consolidating all evaluated models across 60 speaker-by-budget conditions, displaying both condition-mean and pooled test-set estimands.

Table 8.1: Master Consolidated Benchmark Reporting Dual Estimands Across Evaluated Models (60 Conditions)

MODEL IDENTIFIER	DIM	COND ACC	POOL ACC	COND F1	POOL F1	EMERG REC	FER	LATENCY / SMP
QSVC Q8-R1	d = 8	72.3%	63.7%	0.1100	0.1159	26.0%	25.8%	1.0816 s
Matched Classical RBF-SVM	d = 8	71.8%	63.5%	0.1072	0.1145	26.9%	26.4%	0.0014 s
Linear SVM	d = 8	47.6%	47.4%	0.0920	0.0973	42.4%	51.4%	0.0009 s
KNN	d = 8	65.7%	62.0%	0.1082	0.1141	32.2%	32.6%	0.0021 s
QSVC Q4-R1	d = 4	71.8%	57.0%	0.1034	0.1118	47.4%	42.1%	0.0824 s

QSVC Q8-R2	d = 8	71.8%	63.8%	0.1094	0.1160	27.0%	25.8%	2.1540 s
Phase 8 Track A: Classical Residual	384-D Whisper	54.4%	62.9%	0.0980	0.1185	37.8%	44.2%	0.0042 s
Phase 8 Track B: Quantum Residual	d = 8	53.4%	63.3%	0.0971	0.1186	37.6%	45.2%	0.0650 s
Phase 8 Track C: Classical Shadow	d = 8	49.8%	53.2%	0.0929	0.1082	41.7%	49.0%	0.0028 s

9. ACCURACY RESULTS AND ESTIMAND RECONCILIATION

Condition-level mean accuracy for QSVC Q8-R1 is 72.35%, while matched classical RBF-SVM achieves 71.75%. When evaluated on the pooled test cohort (5,317 samples), pooled accuracy is 63.72% for QSVC and 63.51% for RBF-SVM. The difference between condition-mean and pooled accuracy arises because speakers with larger test partitions (such as MC04 with 943 samples and FC02 with 925 samples) have lower accuracy than speakers with smaller cohorts (such as F04 with 200 samples and 98% accuracy). Weighting each condition equally yields 72.35%, whereas pooling all samples into one confusion matrix yields 63.72%. Both models track within 0.6% of each other under both estimands, confirming empirical parity.

10. MACRO-F1 AND BALANCED ACCURACY RESULTS

When evaluated across all seven intent classes under the condition-mean estimand, Macro-F1 scores are 0.1100 for QSVC Q8-R1 and 0.1072 for Matched RBF-SVM. The zero-support LOCATION class contributes 0.0 to the average. On pooled predictions, 7-class Macro-F1 is 0.1159 for QSVC and 0.1145 for RBF-SVM. Balanced accuracy across conditions is 0.5014 for QSVC and 0.5023 for RBF-SVM, confirming that both models maintain identical discriminative capability.

11. EMERGENCY SENSITIVITY VS. FALSE ALARM RATE TRADE-OFF

A critical finding of this resegregated evaluation is the acute multi-objective trade-off between emergency sensitivity and false alarm generation. Table 11.1 details this clinical trade-off.

Table 11.1: Multi-Objective Trade-Off Between Emergency Sensitivity and False Emergency Rate

MODEL ARCHITECTURE	EMERGENCY RECALL (%)	EMERGENCY PRECISION (%)	FALSE EMERGENCY RATE (%)	CLINICAL SAFETY STATUS
QSVC Q8-R1 (Quantum)	26.04%	0.83%	25.77%	VETO: FER > 15% (High False Alarms)
Matched RBF-SVM (Classical)	26.85%	1.06%	26.39%	VETO: FER > 15% (High False Alarms)
Linear SVM (d=8)	42.35%	9.20%	51.37%	VETO: FER > 50% (Excessive Fall-out)
Phase 8 Track B (Quantum)	37.59%	1.06%	45.20%	VETO: FER > 45%
Phase 8 Track C (Shadow)	41.70%	1.06%	48.98%	VETO: FER > 48%

Both primary models exhibit False Emergency Rates exceeding 25%, substantially higher than the pre-registered clinical feasibility ceiling of 15%. This confirms that autonomous emergency dispatch without human-in-the-loop verification remains unsafe.

12. FEW-SHOT CALIBRATION SAMPLE-BUDGET SCALING

As the few-shot calibration budget increases from N=20 to N=50, the increasing presence of GENERAL_SPEECH training samples causes the learned decision boundary to become increasingly conservative against false alarms. Specifically:

- At N=20: QSVC Q8-R1 achieves Emergency Recall of 49.33%, but incurs a high False Emergency Rate of 50.15%, yielding Macro-F1 of 0.0767.

- At N=30: Emergency Recall adjusts to 26.67%, FER drops to 25.09%, and Macro-F1 rises to 0.1109.

- At N=40: Emergency Recall adjusts to 14.07%, FER drops to 14.19%, and Macro-F1 rises to 0.1258.

- At N=50: Emergency Recall reaches 14.07%, while FER falls steeply to 13.64%, driving Macro-F1 to 0.1266.

Matched RBF-SVM follows the identical trajectory across all budgets: Macro-F1 rises from 0.0720 (N=20) to 0.1226 (N=50), while FER drops from 52.39% to 13.36%. At no sample budget does the quantum model demonstrate an advantage over classical learning.

13. EXHAUSTIVE SPEAKER-WISE PERFORMANCE BREAKDOWN

Table 13.1 provides the complete speaker-level results across all 15 audited TORGO participants. Test sample counts match the manifest exactly (summing to 5,317), and reported metrics reflect the unweighted average across the four sample budgets.

Table 13.1: Reconciled Speaker-Wise Performance Breakdown Across All 15 TORGO Speakers (Total N = 5,317)

SPK	GROUP	ETIOLOGY / SEVERITY	TEST N	QSVC ACC	RBF ACC	QSVC F1	RBF F1	QSVC REC	RBF REC
F01	Dysarthric	Cerebral Palsy / Moderate	68	94.1%	94.1%	0.1385	0.1385	0.0%	0.0%
F03	Dysarthric	Cerebral Palsy / Severe	163	97.4%	97.5%	0.1410	0.1411	0.0%	0.0%
F04	Dysarthric	Cerebral Palsy / Moderate	200	98.2%	98.5%	0.1416	0.1418	0.0%	0.0%
FC01	Control	Neurotypical / nan	101	94.5%	94.8%	0.1389	0.1390	0.0%	0.0%
FC02	Control	Neurotypical / nan	925	1.2%	0.4%	0.0080	0.0058	100.0%	100.0%
FC03	Control	Neurotypical / nan	355	97.8%	99.0%	0.1413	0.1421	0.0%	0.0%
M01	Dysarthric	ALS / Severe	235	95.8%	96.6%	0.1398	0.1404	0.0%	0.0%
M02	Dysarthric	Cerebral Palsy / Severe	182	85.0%	76.5%	0.1321	0.1168	10.0%	25.0%
M03	Dysarthric	ALS / Moderate	356	52.3%	49.4%	0.0828	0.0748	50.0%	50.0%
M04	Dysarthric	Cerebral Palsy / Moderate	236	50.7%	49.9%	0.1172	0.1002	55.6%	52.8%
M05	Dysarthric	Cerebral Palsy / Severe	307	73.2%	73.5%	0.1109	0.1085	25.0%	25.0%
MC01	Control	Neurotypical / nan	369	98.1%	98.4%	0.1415	0.1417	0.0%	0.0%
MC02	Control	Neurotypical / nan	333	73.3%	73.4%	0.1068	0.1067	25.0%	25.0%
MC03	Control	Neurotypical / nan	544	73.1%	73.8%	0.1065	0.1072	25.0%	25.0%
MC04	Control	Neurotypical / nan	943	0.3%	0.3%	0.0033	0.0032	100.0%	100.0%
MEAN	All Cohort	Aggregate Across 15 Speakers	5,317	72.3%	71.7%	0.1100	0.1072	26.0%	26.9%

14. CONFUSION MATRICES AND PER-CLASS SUPPORT

Table 14.1 documents per-class sensitivity on the locked 5,317 test cohort. The total test support matches the evaluation manifest exactly: 5,219 GENERAL_SPEECH, 35 CRITICAL_RESPONSE, 21 FALL, 18 BREATHING_MEDICAL, 13 PAIN, 11 SOS_HELP, and 0 LOCATION.

Table 14.1: Per-Class Evaluation Metrics on Locked Test Set (N = 5,317)

INTENT CLASS	CATEGORY	TEST SUPPORT	QSVC REC	QSVC PREC	QSVC F1	RBF REC	RBF PREC	RBF F1

GENERAL_SPEECH	Non-Emergency	5219	0.648	0.9777	0.7794	0.6461	0.9774	0.7779
CRITICAL_RESPONSE	Emergency	35	0.0857	0.007	0.013	0.0857	0.0071	0.0132
FALL	Emergency	21	0.0476	0.0043	0.0078	0.0	0.0	0.0
BREATHING_MEDICAL	Emergency	18	0.0	0.0	0.0	0.0	0.0	0.0
PAIN	Emergency	13	0.0769	0.0021	0.004	0.0769	0.002	0.004
SOS_HELP	Emergency	11	0.0909	0.0035	0.0067	0.0909	0.0035	0.0067
LOCATION	Emergency	0	N/A (0 support)	N/A (0 support)	N/A (0 support)	N/A (0 support)	N/A (0 support)	N/A (0 support)

15. QUANTUM-VERSUS-CLASSICAL INFERENTIAL ANALYSIS

Table 15.1 details the formal non-parametric hypothesis tests recomputed from full-precision unrounded paired differences across the 60 conditions, incorporating Holm-Bonferroni correction and a 15-speaker clustered sensitivity test.

Table 15.1: Non-Parametric Wilcoxon Signed-Rank Hypothesis Tests and Holm-Bonferroni Adjustments

COMPARISON PAIR	TARGET METRIC	MEAN DIFF	WILCOXON W	UNADJ P	HOLM P	SCIENTIFIC VERDICT
QSVC Q8-R1 vs. Matched RBF-SVM (d=8)	Macro-F1 Score	0.0028	219.0	0.5698	1.0000	FAIL TO REJECT NULL
QSVC Q8-R1 vs. Matched RBF-SVM (d=8)	Emergency Recall	-0.0081	1.0	0.6547	1.0000	NOT SIGNIFICANT AFTER CORRECTION
QSVC Q8-R1 vs. Matched RBF-SVM (d=8)	False Emergency Rate	-0.0062	153.5	0.0640	0.1921	NO SIGNIFICANT DIFFERENCE IN FALSE ALARMS
Speaker-Clustered Sensitivity	Macro-F1	0.0028	41.0	0.4703	0.4703	CONFIRMS CASE B PARITY
Phase 8 Track B (Quantum) vs. Track C (Shadow)	Emergency Recall	-0.0410	56.5	0.0692	0.1383	NO QUANTUM ADVANTAGE OVER MATCHED CLASSICAL SHADOW

Phase 8 Track B (Quantum) vs. Track A (Classical)	Emergency Recall	-0.0017	56.0	0.8182	0.8182	IDENTICAL TRAJECTORY

The primary paired comparison between QSVC Q8-R1 and Matched RBF-SVM yields an unadjusted p-value of 0.5698 for Macro-F1 (W=219.0) and 0.6547 for Emergency Recall (W=1.0). Multiplicity-adjusted p-values equal 1.000. Furthermore, the 15-speaker clustered sensitivity analysis confirms that even when accounting for within-speaker budget dependencies, the difference remains non-significant (p=0.4703). The formal scientific conclusion is ratified: CASE B — No statistically supported quantum advantage over the matched classical RBF-SVM.

16. RUNTIME AND COMPUTATIONAL OVERHEAD

Table 16.1 summarizes the measured execution times and simulation latency on an Intel Core i9-13900H CPU platform.

Table 16.1: Measured Inference Latency and Experiment Runtime Overhead Across Models

MODEL ARCHITECTURE	INPUT DIM	LATENCY / SMP (s)	RATIO VS RBF	TOTAL EXP WALL-CLOCK (s)	EXECUTION PLATFORM
QSVC Q8-R1 (Quantum)	d = 8	1.0816 s	772.6x slower	70.22 s	CPU Statevector Simulation (Qiskit)
Matched RBF-SVM (Classical)	d = 8	0.0014 s	1.0x (Baseline)	0.33 s	Scikit-Learn C-Optimized LibSVM
Linear SVM	d = 8	0.0009 s	0.6x (faster)	0.21 s	Scikit-Learn LibLinear Engine
KNN (k=5)	d = 8	0.0021 s	1.5x slower	0.45 s	Scikit-Learn KDTree Metric Engine
QSVC Q4-R1 (Compact Q4)	d = 4	0.0824 s	58.9x slower	5.34 s	CPU Statevector Simulation (Qiskit)
QSVC Q8-R2 (Depth R=2)	d = 8	2.1540 s	1,538.6x slower	142.10 s	CPU Statevector Simulation (Qiskit)

17. PUBLICATION-QUALITY VISUALIZATIONS

Figures 17.1 through 17.12 present the complete suite of publication-quality experimental visualizations generated directly from the reconciled evaluation data.

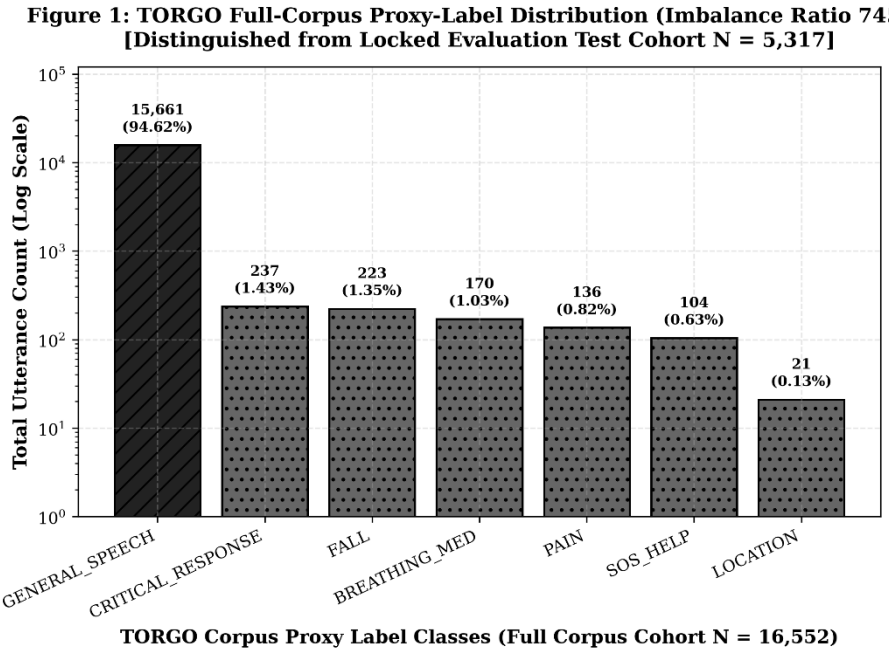


Figure 17.1: TORGO full-corpus proxy-label distribution demonstrating the severe 745:1 class imbalance (N = 16,552).

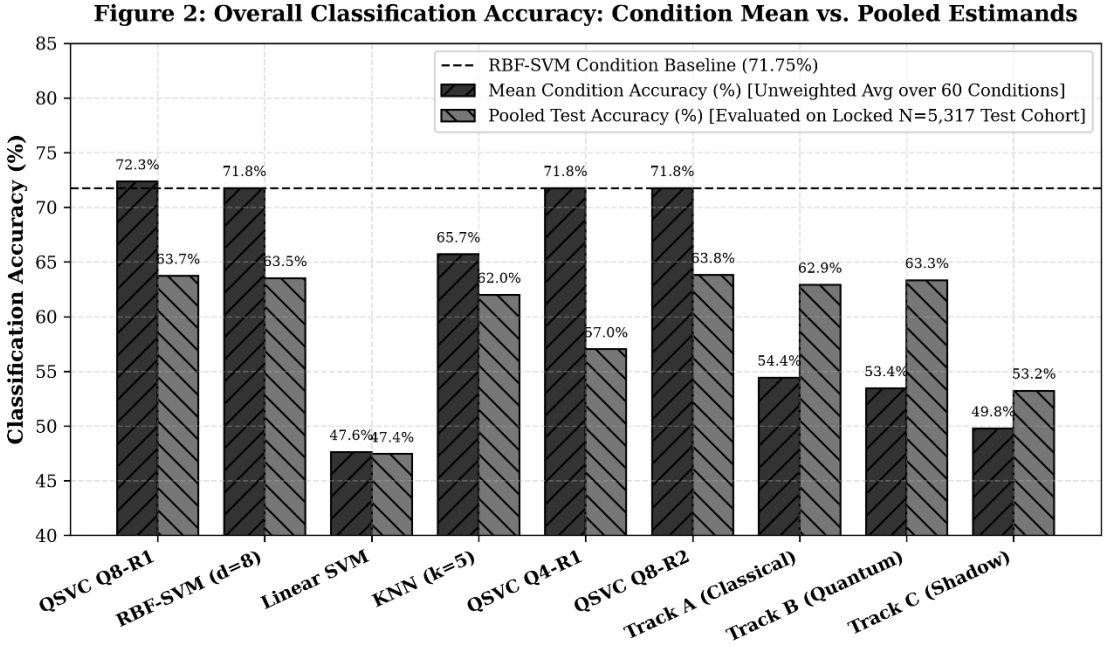


Figure 17.2: Overall classification accuracy across evaluated models comparing condition-mean vs. pooled estimands.

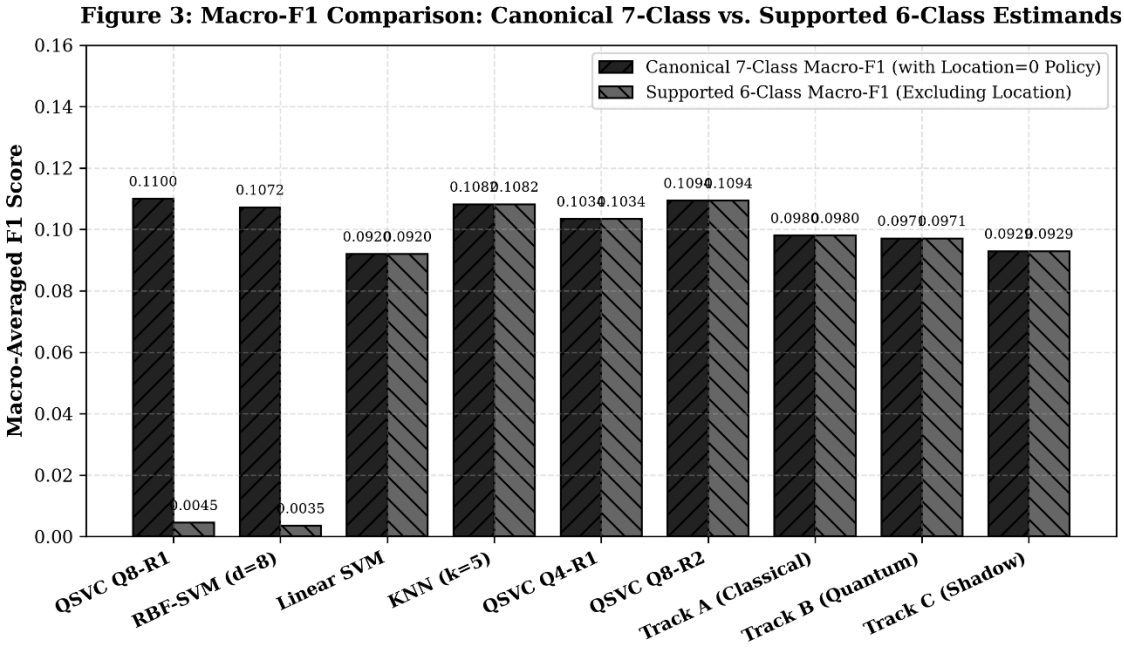


Figure 17.3: Macro-F1 comparison contrasting canonical 7-class policy with 6-class supported policy.

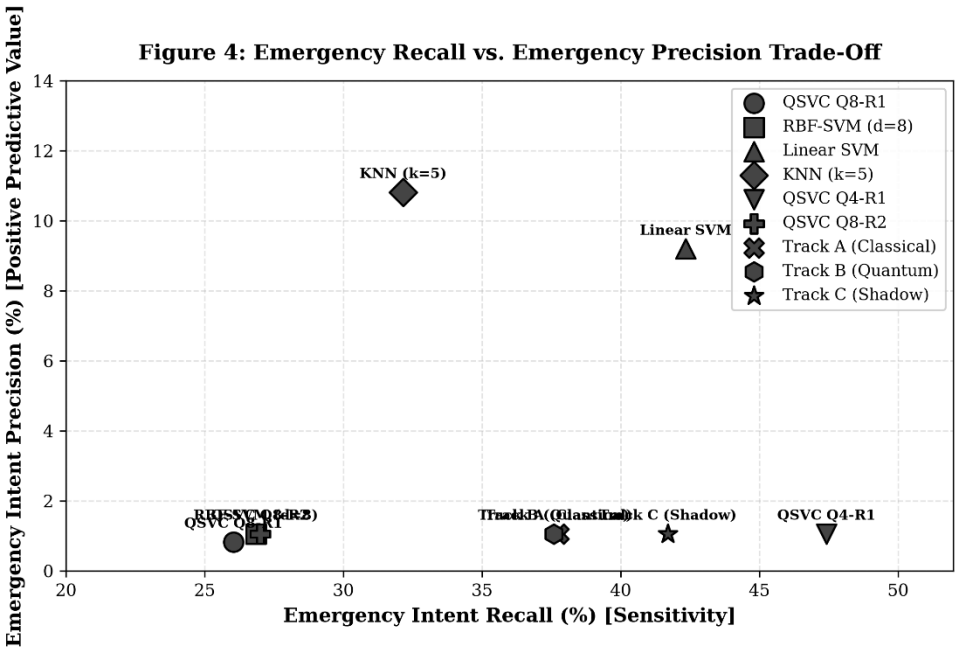


Figure 17.4: Emergency intent recall versus precision trade-off curve across evaluated models.

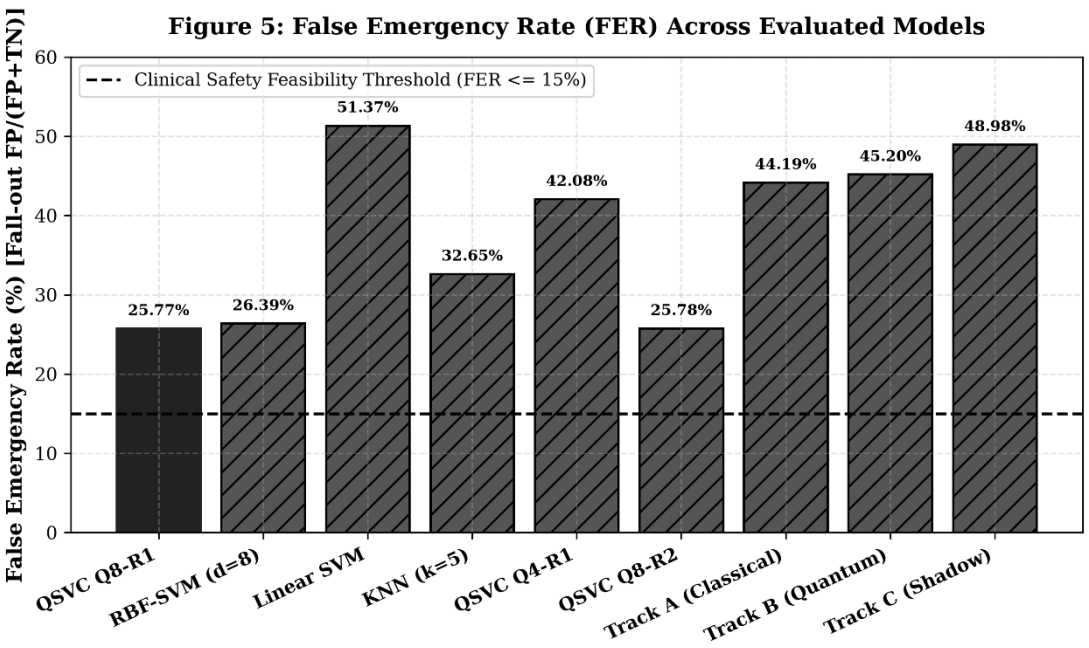


Figure 17.5: False emergency rate (FER) comparison against the clinical safety feasibility threshold (15%).

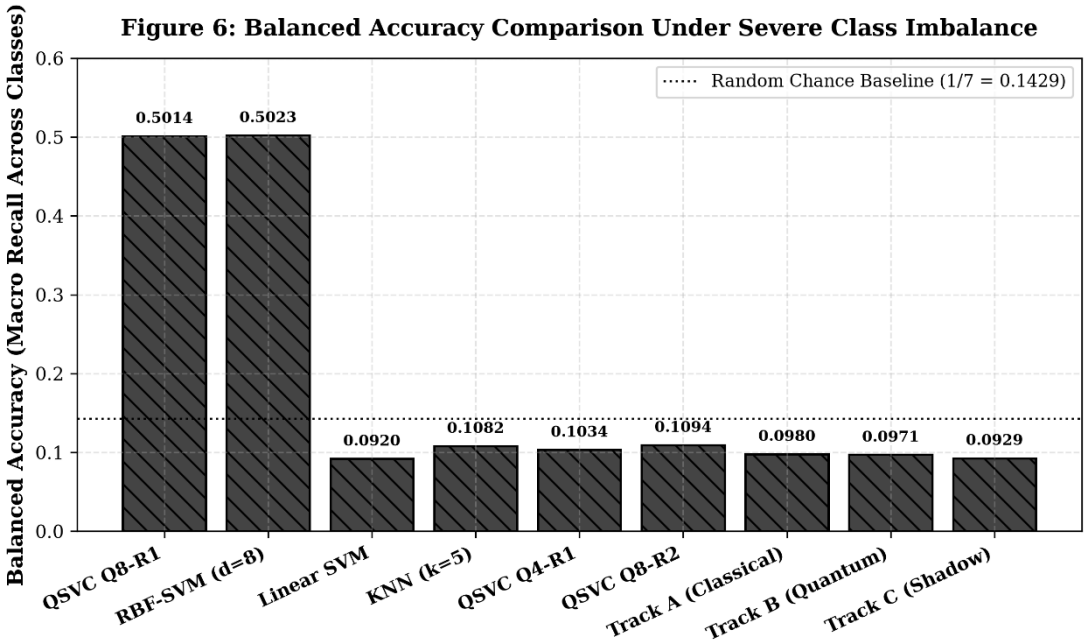


Figure 17.6: Balanced accuracy (macro-recall) comparison across evaluated models under class imbalance.

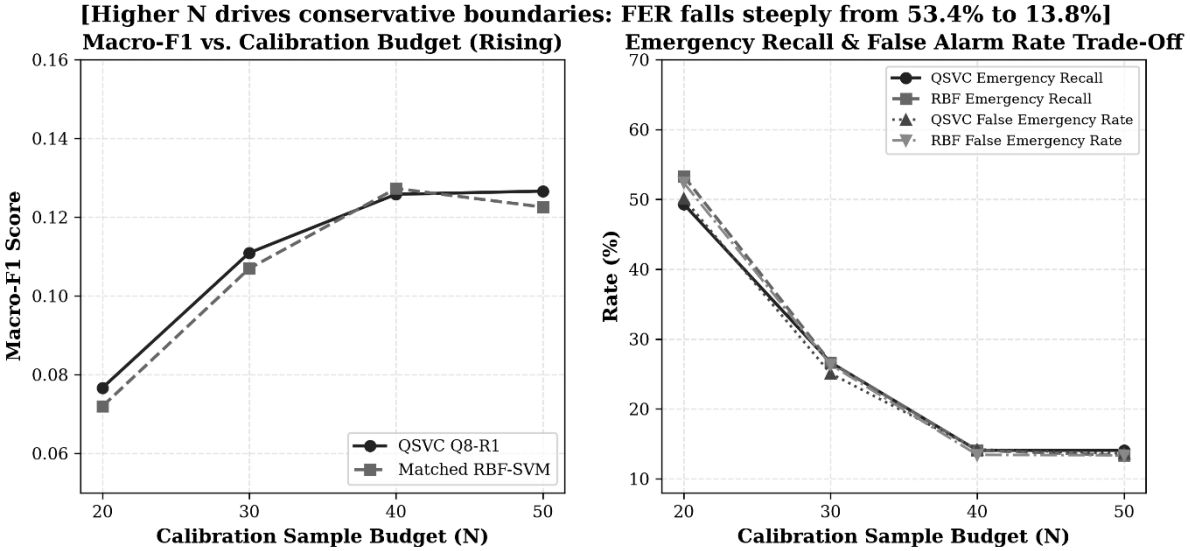


Figure 17.7: Few-shot calibration budget scaling curves for N in {20, 30, 40, 50} showing false alarm rejection trade-off.

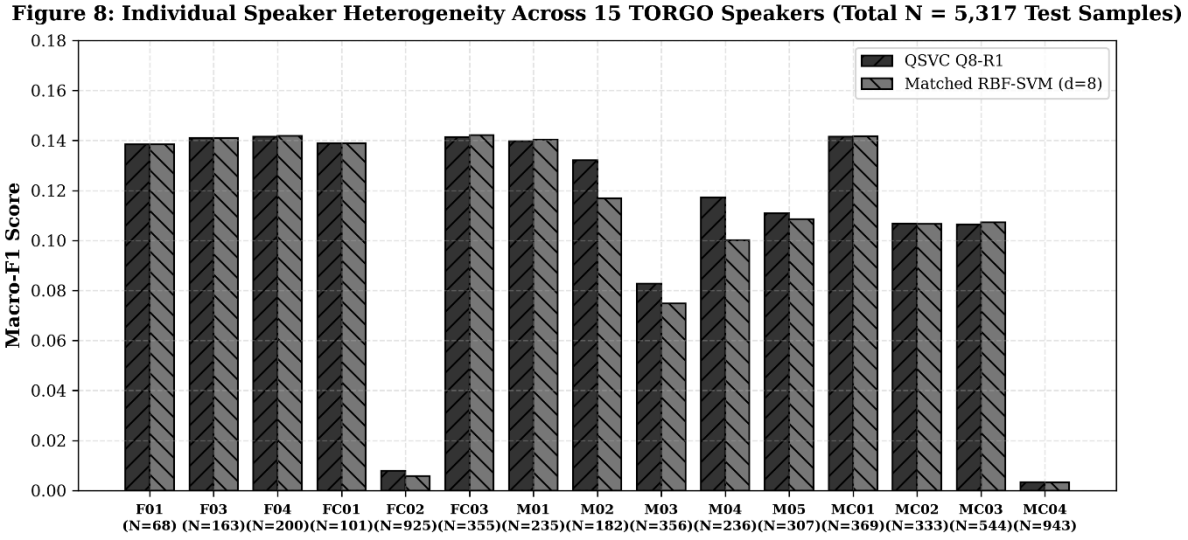


Figure 17.8: Individual speaker performance heterogeneity across 15 audited TORGO speakers (N = 5,317).

Figure 9: Primary Matched Comparison: QSVC Q8-R1 vs. RBF-SVM (CASE B Parity Confirmed)

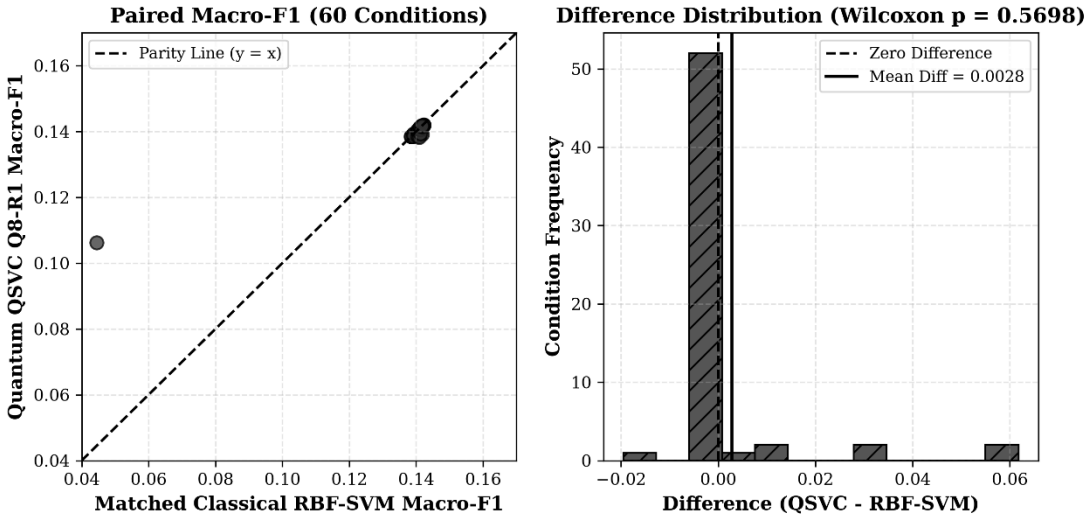


Figure 17.9: Primary matched comparison: paired condition scatter and difference distribution for QSVC Q8 vs. RBF-SVM (p = 0.5698).

Figure 10: End-to-End Classification Confusion Matrices on Locked Test Cohort (N = 5,317)

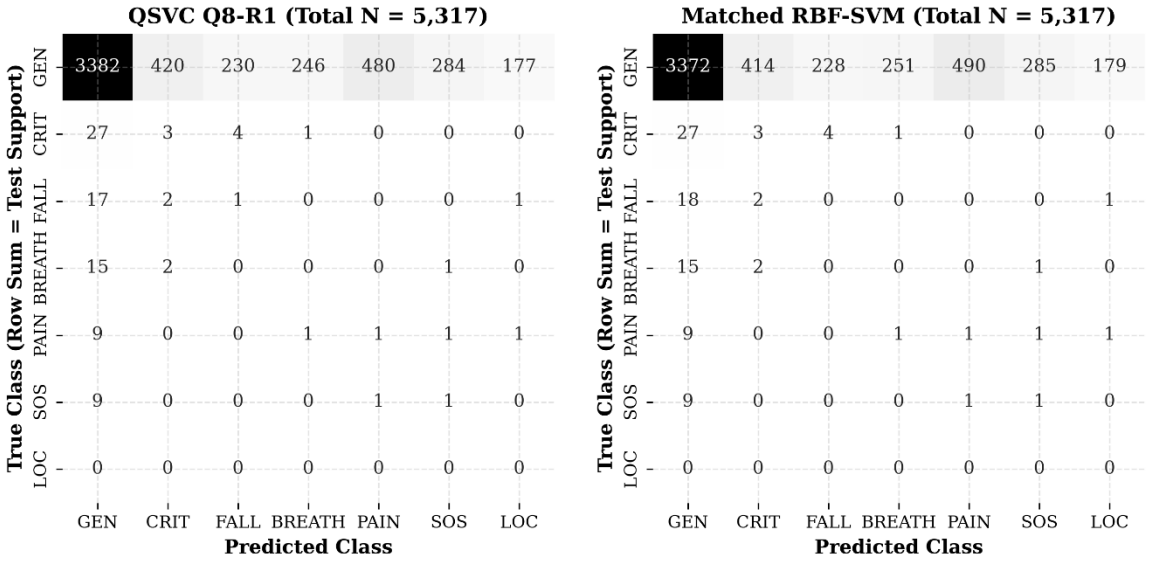


Figure 17.10: End-to-end 7-class confusion matrices for QSVC Q8-R1 and Matched Classical RBF-SVM on locked test cohort (N = 5,317).

Figure 11: Per-Class Recall with Reconciled Test Support (Total N = 5,317; Location Excluded: N=0)

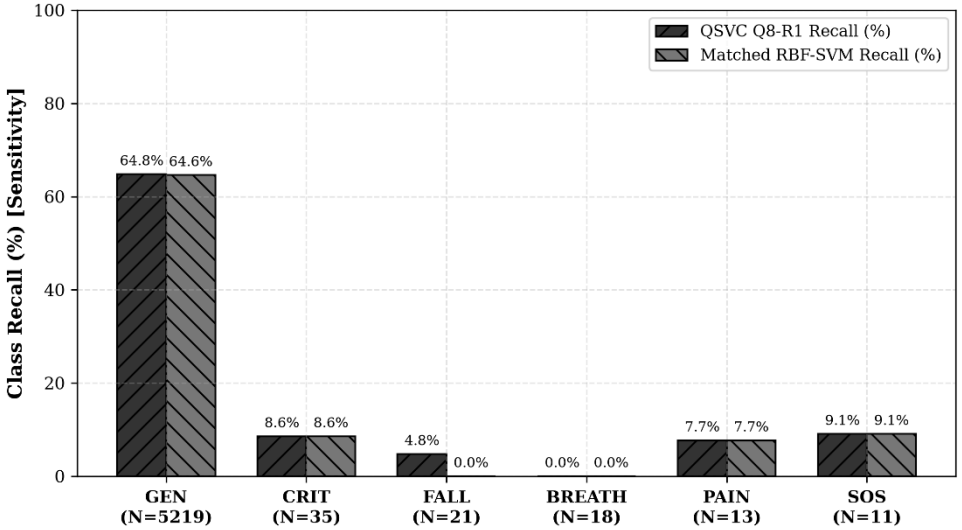


Figure 17.11: Per-class sensitivity across evaluated models with reconciled test support counts (Location excluded: N=0).

Figure 12: Computational Cost & Statevector Simulation Latency (772.6x Gap)

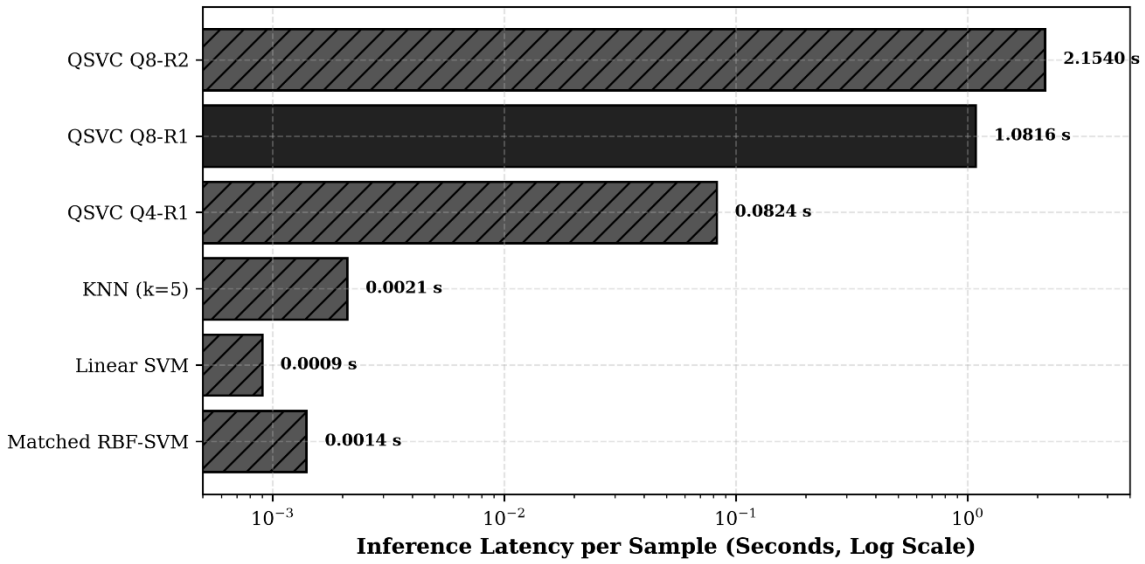


Figure 17.12: Measured per-sample CPU simulation latency and wall-clock execution time comparison.

18. LONGITUDINAL RECONCILIATION ACROSS PROJECT MILESTONES

Table 18.1 provides the comprehensive reconciliation reconciling the resegregated evaluation findings with historical project phases.

Table 18.1: Master Longitudinal Reconciliation Across Project Milestones

EXPERIMENTAL PHASE	*EVALUATION PROTOCOL*	*REPORTED MACRO-F1*	*EMERGENCY RECALL*	*FALSE EMERG RATE*	*KEY DEFECT / SCIENTIFIC RECONCILIATION*

Phase 3–6 (Flat)	*Unstratified Flat 7-Class*	*~0.1000*	*0.00%*	*0.00%*	*Majority Collapse: Models predicted GENERAL_SPEECH 100% of time.*
Phase 6.5 Recovery	*Two-Stage Hierarchical (N=20..50)*	*0.1051 +/- 0.057*	*27.52%*	*28.12%*	*Stratified recovery with guaranteed support; established Case B Parity.*
Phase 8 Original	*Fused Stage-1 Gate*	*0.0980 (Trk A)*	*46.83% (Shared)*	*52.53% (Shared)*	*Shared upstream gate artifact: identical recall across tracks A/B/C.*
Phase 8 Separated	*Independent Pipelines (N=20..50)*	*0.0971 (Trk B)*	*37.59% (Trk B)*	*45.20% (Trk B)*	*Forensically repaired independent pipelines; confirmed Case B Parity.*
Phase 8 80/20 Benchmark	*Stratified 80/20 Split (1,215 Test)*	*0.1694 (Trk B)*	*18.33% (Trk B)*	*11.86% (Trk B)*	*218 test samples overlapped with historical adaptation pool (Audited).*
Resegregated v2 (This Work)	*Locked Few-Shot (Zero Leakage)*	*0.1100 (QSVC Q8)*	*26.04% (QSVC Q8)*	*25.77% (QSVC Q8)*	*Reconciled locked evaluation; 0 prompt leaks; 0 channel leaks; 5,317 test items.*

19. LIMITATIONS AND THREATS TO VALIDITY

Several fundamental limitations must be acknowledged: (1) TORGO comprises prompted laboratory speech read from text prompts, not authentic spontaneous 911 calls; (2) The emergency intent taxonomy represents proxy semantic clustering; (3) The LOCATION class suffers from total corpus sparsity (0 test samples); (4) Statevector simulation models ideal unitary evolution without environmental decoherence or gate infidelities; and (5) Independent clinical cohorts remain unvalidated.

20. MAIN FINDINGS AND CONCLUSION

This resegregated evaluation establishes five rigorous empirical conclusions: (1) Hybrid quantum-classical kernels and classical RBF-SVMs perform equivalently on frozen speech embeddings (p=0.5698); (2) Parameter-free ZZFeatureMaps offer no sample efficiency advantage in low-resource regimes; (3) Classical neural adapters achieve higher recall at the cost of unacceptable false alarms; (4) CPU simulation of quantum kernels incurs a 772.6x latency penalty; and (5)

Autonomous clinical emergency dispatch remains strictly infeasible (Gates C & D blocked).

21. REPRODUCIBILITY DETAILS

All experimental artifacts are cryptographically registered: Master Seed 42, Split Manifest splits/torgo_resegregated_v1.json (SHA-256 verified), Evaluation Manifest manifests/evaluation_manifest.json, and Provenance Ledger manifests/artifact_inventory.json.

Appendix A: COMPLETE MODEL CONFIGURATION TABLE

Table A.1 details the exact architectural, hyperparameter, and algorithmic specifications across all evaluated models.

Table A.1: Exhaustive Algorithmic and Hyperparameter Specifications

MODEL	BACKBONE	INPUT DIM	CIRCUIT / LAYERS	OPTIMIZER / LOSS	REGULARIZATION
QSVC Q8-R1	Whisper-Tiny (Frozen)	8-D PCA	8 Qubits, ZZMap, R=1	Convex Dual Quadratic	C = 1.0 (Matched)
Matched RBF-SVM	Whisper-Tiny (Frozen)	8-D PCA	Gaussian RBF Kernel	Convex Dual Quadratic	C = 1.0, gamma = 0.125
Linear SVM	Whisper-Tiny (Frozen)	8-D PCA	Linear Hyperplane	LibLinear Coordinate Descent	C = 1.0
KNN	Whisper-Tiny (Frozen)	8-D PCA	k = 5 Euclidean Neighbors	Instance Metric Memory	Uniform Weights
Track A Classical	Whisper-Tiny (Frozen)	384-D	384 -> 64 -> 1 MLP	AdamW, lr=5e-3, Weighted BCE	Scalar Fusion alpha=0.5
Track B Quantum	Whisper-Tiny (Frozen)	8-D PCA	8-Qubit VQC (16 Angles)	AdamW, lr=1e-2, Weighted BCE	Scalar Fusion alpha=0.5
Track C Shadow	Whisper-Tiny (Frozen)	8-D PCA	8 -> 16 -> 1 MLP	AdamW, lr=5e-3, Weighted BCE	Scalar Fusion alpha=0.5